

Homework 2

Syed N. Asad Ali (sa09475)

Q.1) 2.1

Let $x(t) = \cos(\omega_0 t)$ from $[-T/2, T/2]$
 using Euler's identity

$$X(\omega) = \int_{-T/2}^{T/2} \cos(\omega_0 t) e^{-j\omega t} dt$$

$$= \frac{1}{2} \int_{-T/2}^{T/2} e^{-j(\omega - \omega_0)t} dt + \int_{-T/2}^{T/2} e^{-j(\omega + \omega_0)t} dt$$

$$\therefore \int_{-T/2}^{T/2} e^{-j\omega t} dt = \frac{2 \sin(aT/2)}{a}$$

$$X(\omega) = \frac{\sin((\omega - \omega_0)T/2)}{\omega - \omega_0} + \frac{\sin((\omega + \omega_0)T/2)}{\omega + \omega_0}$$

If the window spans an integer Number N of periods,

$$T = N \cdot \frac{2\pi}{\omega_0}$$

$\rightarrow \omega_0 \cdot T/2 = N\pi$; Then:

$$X(0) = 0 \text{ (zero } N()) \text{ since } \sin(N\pi) = 0$$

peaks at $\omega = \pm \omega_0$ with value $X(\pm \omega_0) = T/2$

zeros at $\omega = \pm \omega_0 + 2\pi k/T$ for Non-zero integers.

Q.2) 2.2

$$\text{Let } x(t) = e^{-t/T} \sin(2\pi f_0 t) u(t) \quad T > 0$$

$$X(f) = \int_0^{\infty} e^{-t/T} \sin(2\pi f_0 t) e^{-j2\pi f t} dt$$

using,

$$\sin(2\pi f_0 t) = \frac{1}{2j} (e^{j2\pi f_0 t} - e^{-j2\pi f_0 t})$$

$$\int_0^{\infty} e^{-(a+j2\pi v)t} dt = \frac{1}{a+j2\pi v} \quad \text{for } a > 0$$

$$X(f) = \frac{1}{2j} \left[\frac{1}{1/T + j2\pi(f-f_0)} - \frac{1}{1/T + j2\pi(f+f_0)} \right]$$

$$\text{Poles are at } f = \pm f_0 - \frac{j}{2\pi T}$$

Q.3) 2.3

$$(a) \quad X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt, \quad x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi f t} df$$

$$\sum_{m=-\infty}^{\infty} x(m) = \sum_{k=-\infty}^{\infty} X(k)$$

If we take $T=1$ in the general sampling relation we get:

$$\sum_{n=-\infty}^{\infty} x(n) e^{-j2\pi n f} = \sum_{k=-\infty}^{\infty} X(f-k)$$

$$\text{setting } f=0 \text{ gives: } \sum_{m=-\infty}^{\infty} x(m) = \sum_{k=-\infty}^{\infty} X(k)$$

$$b) \sum_{m=-\infty}^{\infty} x(mT) = \frac{1}{T} \sum_{n=-\infty}^{\infty} X(n/T)$$

$$x_s(t) = \sum_{m=-\infty}^{\infty} x(mT) \delta(t - mT)$$

$$= \frac{1}{T} \sum_{n=-\infty}^{\infty} \delta(f - n/T)$$

using multiplication convolution property

$$X_s(f) = X(f) \cdot \frac{1}{T} \sum_{n=-\infty}^{\infty} \delta(f - n/T)$$

$$\frac{1}{T} \sum_{n=-\infty}^{\infty} X(f - n/T)$$

train

∴ Transformation from impulse gives :

$$X_s(f) = \sum_{m=-\infty}^{\infty} x(mT) e^{-j2\pi f m T}$$

∴ setting $f=0$ and equating both gives

$$\sum_{m=-\infty}^{\infty} x(mT) = \frac{1}{T} \sum_{n=-\infty}^{\infty} X(n/T)$$

$$c) \sum_{n=-\infty}^{\infty} x(nT) e^{-j2\pi n f T} = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(f - k/T)$$

$$X_s(f) = X(f) \cdot \frac{1}{T} \sum_{k=-\infty}^{\infty} \delta(f - k/T) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(f - k/T)$$

$$X_s(f) = \sum_{n=-\infty}^{\infty} x(nT) e^{-j2\pi n f T}$$

$$\sum_{n=-\infty}^{\infty} x(nT) e^{-j2\pi n f T} = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(f - k/T)$$

Q.4) 2.4

$$x(t) = \Lambda(t) = \begin{cases} 1 - |t| & , |t| \leq 1 \\ 0 & , \text{otherwise (repeat with } T=2) \end{cases}$$

even function $\rightarrow$ only cosine terms in its fourier series
with $\omega_0 = \frac{2\pi}{T} = \pi \text{ rad/s}$.

co-efficients:

$$a_0 = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cdot dt = \frac{2}{2} \int_{-1}^1 (1 - |t|) \cdot dt = 1 \Rightarrow \frac{a_0}{2} = \frac{1}{2}$$

for $k \geq 1$:

$$a_k = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos(k\pi t) \cdot dt = \frac{4}{2} \int_0^1 (1-t) \cos(k\pi t) \cdot dt$$

$$= 2 \left[\int_0^1 \cos(k\pi t) \cdot dt - \int_0^1 t \cos(k\pi t) \cdot dt \right]$$

$$a_k = \frac{-2 \cdot (-1)^k - 1}{(k\pi)^2} = \frac{2(1 - (-1)^k)}{k^2 \pi^2}$$

$$a_k = 0 \text{ for even } k$$

$$a_k = \frac{4}{k^2 \pi^2} \text{ for odd } k$$

$$x(t) = \frac{1}{2} \sum_{\substack{k=1 \\ k=\text{odd}}}^{\infty} \frac{4}{k^2 \pi^2} \cos(k\pi t) = \frac{1}{2} + \frac{4}{\pi^2} \left[\frac{\cos(\pi t)}{3^2} + \frac{\cos(5\pi t)}{5^2} + \dots \right]$$

$\Rightarrow$ Only odd harmonics appear.

Q.5) 2.6

$$H(f) = e^{-2\pi^2 f^2 r^2}$$

(a)

3-dB Bandwidth

$$H(f_{3dB}) = e^{-2\pi^2 f_{3dB}^2 r^2} = \frac{1}{\sqrt{2}}$$

$$\therefore -2\pi^2 f_{3dB}^2 r^2 = \ln\left(\frac{1}{\sqrt{2}}\right) = -\frac{1}{2} \ln(2)$$

$$f_{3dB}^2 = \frac{\ln 2}{4\pi^2 r^2}$$

$$f_{3dB} = \frac{\sqrt{\ln 2}}{2\pi r}$$

(b)

$$x(t) = 3e^{-\pi(10t)^2}$$

The Fourier transform of a Gaussian $e^{-\pi at^2}$ is another Gaussian.

$$\mathcal{F}\{e^{-\pi at^2}\} = \frac{1}{\sqrt{a}} e^{-\pi f^2/a}$$

$$a = 100; \quad X(f) = \frac{3}{10} e^{-\pi f^2/100}$$

Output Spectrum:

$$Y(f) = H(f) X(f)$$

$$H(f) = e^{-\pi(f/5)^2} = e^{-\pi f^2/25}$$

$$Y(f) = \frac{3}{10} e^{-\pi f^2/100} \cdot e^{-\pi f^2/25} = \frac{3}{10} e^{-\pi f^2 \left(\frac{1}{100} + \frac{1}{25}\right)}$$

$$Y(f) = \frac{3}{10} e^{-\pi f^2/20}$$

$$S_y(f) = |Y(f)|^2$$

$$S_y(f) = (3/10)^2 e^{-2\pi f^2/20}$$

$$S_y(f) = \frac{9}{100} e^{-\pi f^2/10}$$

Q.6) 2.7

$$\Pi(t) = \begin{cases} 1, & |t| < 1/2 \\ 0, & \text{otherwise} \end{cases}$$

$$\phi(t) = \lim_{i \rightarrow \infty} \Pi(t) * 2\Pi(2t) * 4\Pi(4t) \dots 2^i \Pi(2^i t)$$

$$X(f) = \int x(t) e^{-j2\pi f t} dt$$

$$\bullet \Pi(t) \xrightarrow{f} \text{sinc}(f)$$

$$\bullet \text{Time scaling: } x(at) \xrightarrow{f} \frac{1}{|a|} X\left(\frac{f}{a}\right)$$

$$2^k \Pi(2^k t) \xrightarrow{f} \text{sinc}\left(\frac{f}{2^k}\right)$$

$$\bullet \text{Convolution in time} \longleftrightarrow \text{multiplication in frequency hence,}$$

$$\phi(f) = \prod_{k=0}^{\infty} \text{sinc}\left(f/2^k\right)$$

$$\text{support of } \phi(t): \sum_{k=0}^{\infty} \frac{1}{2^k} = 2 \rightarrow \text{supp } \phi(t) = [-1, 1]$$

Area under $\phi(t)$

$$\int_{-\infty}^{\infty} 2^k \Pi(2^k t) dt = 1 \quad \forall k$$

Thus the area under any finite convolution is 1 and the limit remains 1.

$$\int_{-\infty}^{\infty} \phi(t) \cdot dt = 1 \approx \phi(0) = 1$$

Function equation for $\phi(t)$

$$\phi(f) = \text{sinc}(f) \prod_{k=1}^{\infty} \text{sinc}(f/2^k) = \text{sinc}(f) \phi(f/2)$$

$X(f/2) \xleftrightarrow{f} 2x(2t)$ and Product $\leftrightarrow$ convolution.

$$\phi(t) = \sqrt{t}(t) * [2\phi(2t)]$$

which is equation $\phi(t)$ must satisfy.

Q.7) 2-8

$$a = 1$$

$$20^2$$

$$X(f) = \sqrt{\frac{\pi}{a}} e^{-(\pi f)^2/a} = \sqrt{2\pi 6^2} e^{-2\pi^2 6^2 f^2}$$

$$X(f) \propto e^{-2\pi^2 6^2 f^2}$$

$$|X(f)| = (e^{-2\pi^2 6^2 f^2})$$

$$\therefore 2\pi^2 6^2 f^2 = f^2$$

$$2\pi^2 6^2 f$$

$$\therefore 4\pi^2 6^2 6^2 f = 1$$

$$\therefore 6f = \frac{1}{2\pi 6}$$

$66f = \frac{1}{2\pi}$

$\Rightarrow$ Gaussian pulse achieves the minimum possible product of time and frequency spreads.